

Performance Testing

Date	06 July 2026
Team ID	
Project Name	Smart Lender — Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual timing via browser DevTools Network tab (no load-testing tool used given project scope)
Type of Testing	Functional + manual response-time observation
Target Module / API	/submit route (form submission to prediction result)
Test Environment	Local (Anaconda Flask development server)
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit a low-risk applicant profile	1	1	Returns 'Loan Will be Approved'
2	Submit a high-risk applicant profile	1	1	Returns 'Loan Will Not be Approved'
3	Submit form with all required fields left at default	1	1	Browser-side validation blocks submission
4	Reload home page repeatedly	1	1	Page and static assets load without error each time

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	~0.3 seconds	Pass	Single-user local test
2	Response Time (Max)	< 5 seconds	~0.6 seconds	Pass	Includes model load + scaling + prediction
3	Throughput (Req/sec)	N/A (single-user scope)	N/A	N/A	Load testing tool not used in this iteration
4	Error Rate	< 1%	0%	Pass	No errors across repeated manual test submissions
5	CPU Utilization	< 80%	Not measured	N/A	Not profiled — local laptop testing only
6	Memory Utilization	< 80%	Not measured	N/A	Not profiled — local laptop testing only

Step 4: Observations & Analysis

Since Smart Lender is a single-user, form-driven prediction tool rather than a high-traffic service, formal load testing (e.g. with JMeter or Locust) was out of scope for this iteration. Response times were observed manually and were well within acceptable limits for an interactive web form. Future iterations intended for production deployment should include concurrent-user load testing before going live.

Step 5: Screenshots / Evidence

UNDERWRITING DECISION

Decision rendered by XGBoost model · Smart Lender Credit Engine

APPROVED

Loan Will be Approved

Based on the applicant's profile, the model assesses this loan as a low credit risk and recommends proceeding to manual sign-off.

Name	Status	Type	Initiator	Size	Time
submit	200	docume...	Other	4.8 kB	134 ms
css2?family=Fraunces:opsz,wgh...	200	stylesheet	submit:8	1.0 kB	133 ms
style.css	304	stylesheet	submit:9	0.3 kB	13 ms
6NU78FyLNQOQZAnv9bYEvDil...	200	font	fonts.googleapis.c...	(memor...)	0 ms
UcC73FwrK3iTeHuS_nVMrMxC...	200	font	fonts.googleapis.c...	(memor...)	0 ms
tDbv2o-fIEEnyOFZhsfku5WU4zr...	200	font	fonts.googleapis.c...	(memor...)	0 ms

LOAN APPLICATION

Applicant Details REF 42JB0M

GENDER

MARRIED

DEPENDENTS 0-3

EDUCATION

SELF EMPLOYED

Name	Status	Type	Initiator	Size	Time
predict	200	docume...	Other	8.0 kB	17 ms
style.css	304	stylesheet	predict:9	0.3 kB	12 ms
css2?family=Fraunces:opsz,wgh...	200	stylesheet	predict:8	(memor...)	0 ms
6NU78FyLNQOQZAnv9bYEvDil...	200	font	css2?family=Fraun...	(memor...)	0 ms
UcC73FwrK3iTeHuS_nVMrMxC...	200	font	css2?family=Fraun...	(memor...)	0 ms
tDbv2o-fIEEnyOFZhsfku5WU4zr...	200	font	css2?family=Fraun...	(memor...)	0 ms